

claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-inde  
xer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-  
a5480e64679396293.jsonl`  
- SHA-256: `aa74e256f14c6642ce887bdf4e886027f0b927e3efa84ff3c0e27f7b5be5d30d`  
- Source modified: `2026-06-06T17:43:42+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_6237b110-d58`  
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`
```

Transcript

1. user (2026-06-06T17:41:32.729Z)

Synthesis: research report

****Question:**** Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):

- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:

1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ? especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?
5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio).

19 claims survived 3-vote adversarial verification. Merge semantic duplicates and synthesize.

Confirmed claims

[0] The paper detects rally boundaries using only court geometry and player pose features (not audio or shuttle-detector trajectory): a

Vote: 2-1 · Source: https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15 (primary)

Quote: "A technique has been presented to detect the start of service and end of rally that utilizes court details and poses to accurately identify the service action from other badminton actions... Gradients are used to detect the flight of the shuttlecock till it eventually comes to rest which signifies the end of the rally."

Verifier evidence (high): CONFIRMED by the primary source (Springer chapter

10.1007/978-3-031-69073-0_15, "Service and End of Rally Detection in Badminton Videos") and an independent extended journal version (SN Computer Science 10.1007/s42979-025-03935-0). Every substantive technical claim checks out:

1. START via court + pose, classified by CNN/BiLSTM ? DIRECTLY SUPPORTED. Quote: "A technique has been developed to detect the start of service and end of rally, utilizing court details and poses to accurately identify the service action from other badminton actions." And: "For detecting the start of service, two different models?Bidirectional Long Short Term Memory (BiLSTM) and Convolutional Neural Network (CNN)?were used." Inputs are "12 frames with 26 features from each frame comprising distance, angle, gradient and movements... typical of serving and receiving actions" ? i.e., pose/court-geometry features, not a TrackNet shuttle trajectory. BiLSTM ~95% accuracy (frame interval 4); CNN similar (interval 5).

2. END via shuttlecock-flight gradient analysis ? DIRECTLY SUPPORTED. Quote: "Gradients are used to detect the flight of the shuttlecock till it eventually comes to rest which signifies the end of the rally." The algorithm uses missing/out-of-view shuttle coordinates and requires ">=80% of consecutive frames exhibit no motion of the shuttlecock" to mark rally end (~95% over 187 rallies).

3. No audio ? consistent: all features are spatial (distance/angle/gradient/movement from pose+court+shuttle position); audio is never mentioned in any methodology description.

NUANCE (not refutation): the claim's parenthetical "(not ... shuttle-detector trajectory)" is slightly loose, because rally END explicitly relies on shuttlecock flight COORDINATES (a form of shuttle tracking/trajectory). However the claim itself immediately states END uses "shuttlecock-flight gradient analysis," so the parenthetical is best read as scoping the START mechanism only ? and START genuinely uses no trajectory and no audio. The core methodological assertions are all accurate and primary-sourced.

[1] The service-detection input is hand-engineered pose/geometry features (no raw frames): 12 frames x 26 features per frame (distan

Vote: 3-0 · Source: https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15 (primary)

Quote: "A total of 12 frames with 26 features from each frame comprising distance, angle, gradient and movements typical of serving and receiving actions was arranged in a 2D format that was used as inputs to the models."

Verifier evidence (high): The claim is well-supported by the primary source and corroborated across multiple independent sources. The Springer book chapter (DOI

10.1007/978-3-031-69073-0_15) and its extended journal version (SN Computer Science 6(5):447, 2025) both state verbatim: "A total of 12 frames with 26 features from each frame comprising the distance, angle, gradient and movements that are typical of serving and receiving actions was arranged in a 2D format that was used as inputs to the models," and that the models are BiLSTM and CNN (gradient computation handles rally end). This directly confirms every concrete component of the claim: 12 frames, 26 features/frame, the four feature types (distance/angle/gradient/movement), the 2D arrangement, and the BiLSTM + CNN models.

The one interpretive element ? "hand-engineered pose/geometry features (no raw frames)" ? is a justified gloss rather than a misread. The features are explicitly enumerated as scalar geometric/kinematic quantities (distance, angle, gradient, movement) of player serving/receiving actions, forming a 12x26 numeric matrix. That is, by definition, a hand-engineered feature representation, not pixel/raw-frame input (raw frames would be described as image tensors, not "26 features comprising distance, angle"). The "(no raw frames)" parenthetical is therefore a correct logical consequence, and "pose/geometry" accurately characterizes distance/angle features of player actions. The abstract does not literally use the word "pose"/"skeleton," so that label is mild inference, but it does not overstate the source. Could not access full-text methodology (Springer paywalled; ResearchGate 403) to confirm the exact keypoint source, but no

contradicting evidence exists and the feature description rules out raw-frame input.

Source quality is sufficient: peer-reviewed Springer primary source plus a 2025 journal extension; recent (not outdated); not marketing. No source disputes or qualifies the description.

[2] Badminton point/rally segmentation can be done by a per-frame CNN image classifier that distinguishes broadcast-camera rally fra

Vote: 3-0 · Source: <https://arxiv.org/pdf/2308.12199> (primary)

Quote: "Rally frames are usually always shot from the broadcast camera view whereas the non-rally frames can include close-ups. ... CNN: F1-Score 97.23, Precision 98.06, Recall 92.34"

Verifier evidence (high): VERIFIED against the primary source (arXiv 2308.12199, "Towards Real-Time Analysis of Broadcast Badminton Videos"; I downloaded the PDF and extracted text). Every element of the claim is faithfully reproduced: (a) Method ? "we use a convolutional neural network based classifier to classify rally and non-rally frames. We finetune the ResNet-18 model (pre-trained on ImageNet)" (Sec IV-A). (b) Framing ? verbatim: "Rally frames are usually always shot from the broadcast camera view whereas the non-rally frames can include close-ups of players, additional graphics (score cards) and unusual camera angles." (c) Metrics ? Table II's CNN column is exactly F1 97.23 / Precision 98.06 / Recall 92.34 (the other column, HOG+SVM from Ghosh et al., is 95.44/97.83/91.02). (d) "Without shuttle tracking" ? TRUE: the pipeline is Point-Segmentation(CNN) -> YOLOv3 player localization -> homography -> distance; no shuttle trajectory anywhere. (e) Dataset = 23,156 annotated frames (15,147 non-rally / 8,009 rally). Source is primary and field-trialed (PBL 2019); no contradicting source found. TWO IMPORTANT SCOPE QUALIFICATIONS the claim's wording understates but does not falsify: (1) 97.23 is PER-FRAME binary-classification F1, NOT temporal start/end (IoU) segmentation F1 ? the paper has NO IoU/boundary/edit-score metric (confirmed absent); actual rally boundaries come from a hand-tuned heuristic ("a rally is at least 3 seconds long ... min 3 seconds gap"). So "rally segmentation achieving F1 97.23" conflates frame-classification accuracy with boundary-localization accuracy, which is precisely the IoU-0.5 metric our project struggles with. (2) The discriminative signal is BROADCAST-specific (replays, score-card graphics, director cuts, close-ups); it will NOT transfer to single-fixed-camera amateur/self-recorded footage that has no replays or cuts ? there the classifier has nothing to separate. These are scope caveats relevant to our local-first context, not refutations of the cited fact. The literal claim is well-supported by a current primary source.

[3] This rally segmentation method imposes the same heuristic temporal constraints the user's pipeline uses (minimum rally duration

Vote: 3-0 · Source: <https://arxiv.org/pdf/2308.12199> (primary)

Quote: "A rally is at least 3 seconds long and there is a minimum of 3 seconds gap between two rallies."

Verifier evidence (high): Verified against the primary source (arXiv 2308.12199, "Towards Real-Time Analysis of Broadcast Badminton Videos," Nilesh/Sharma/Ghosh/Jawahar, 2023). Section IV-A (Point Segmentation) states verbatim: "We assume that a rally is at least 3 seconds long and there is a minimum of 3 seconds gap between two rallies. This allows us to group frames of the match into individual rallies by separating sets of continuous rally frames with at least 3 seconds of non-rally frames in between them." This confirms (a) minimum rally duration = 3s, (b) minimum inter-rally gap = 3s, and (c) both are heuristic post-processing constraints applied to CNN frame-level rally/non-rally classifications to segment rallies ? exactly mirroring the user's pipeline (min-duration + merge-gap heuristics). The claim's supporting quote is accurate; it merely drops the "We assume that" prefix without distorting meaning. No contradicting source found; the body text was confirmed via the ar5iv rendering of the same arXiv paper. Source quality (academic paper from C.V. Jawahar's CVIT/IIIT-H group) matches the strength of this low-stakes methodological restatement. Minor caveat for synthesis: these are heuristic assumptions chosen by the authors, not learned/validated optima ? but the claim only asserts the constraints exist and match the user's, which is true.

[4] ShuttleSet is a human-annotated stroke-level badminton singles dataset containing 104 sets, 3,685 rallies, and 36,492 strokes across

Vote: 3-0 · Source: <https://arxiv.org/abs/2306.04948> (primary)

Quote: "104 sets, 3,685 rallies, and 36,492 strokes in 44 matches between 2018 and 2021"

Verifier evidence (high): The claim is fully supported by the primary source (arXiv 2306.04948, also published peer-reviewed at KDD '23, ACM SIGKDD). The arXiv abstract states verbatim: "104 sets, 3,685 rallies, and 36,492 strokes in 44 matches between 2018 and 2021" and "shot type with a choice of 18 distinct classes, the corresponding hitting locations, and the locations of both players at each stroke." The paper title confirms "A Human-Annotated Stroke-Level Singles Dataset for Badminton." Every numeric figure in the claim (104/3,685/36,492/44, 2018-2021, 18 shot classes, hitting locations, both players' locations) matches the abstract exactly.

Independent sources (DeepAI, ResearchGate, ACM DL [dl.acm.org/doi/10.1145/3580305.3599906](https://doi.org/10.1145/3580305.3599906), ar5iv full text) reproduce identical numbers. No source disputes or qualifies the figures. The only nuance is that the abstract adds "27 top-ranking men's and women's singles players," which the claim omits but does not contradict. Source quality (primary arXiv + peer-reviewed KDD '23) matches the descriptive, factual nature of the claim; not marketing or cherry-picked benchmark.

[5] An ML classifier on acoustic waveforms (XGBoost) detects tennis ball-racket impacts with median per-fold accuracy 0.98 (range 0.96-0.99)

Vote: 2-1 · Source: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11843912/> (primary)

Quote: "accuracy across each fold ranged between 0.96 and 0.99, with a median value of 0.98"

Verifier evidence (high): The claim's numbers are EXACTLY and faithfully quoted from the primary peer-reviewed source (Manfredi et al., "Tennis Timing Assessment by a Machine Learning-Based Acoustic Detection System: A Pilot Study," MDPI J. Funct. Morphol. Kinesiol. 2025 / PMC11843912). Verbatim: "the accuracy across each fold ranged between 0.96 and 0.99, with a median value of 0.98." The XGBoost model is indeed the binary impact-vs-noise classifier.

Critically, the claim's qualifier "in isolation" is load-bearing and CORRECT ? it is exactly the right framing. The 0.98 is the accuracy of the standalone classifier stage on 1,400 PRE-SEGMENTED isolated audio samples (509 noise / 442 rebound / 448 impact) under 5-fold CV (1120 train / 280 test per fold), a 2-class problem (impact vs noise). It is NOT end-to-end detection on continuous match audio. The same paper reports the full on-court pipeline drops to 85% average accuracy (and as low as 70.83% for rebounds in group G2), and a separate rebound classifier scores 0.97. The paper reports only accuracy + Cohen's kappa ? no precision/recall, no confusion matrices, and the upstream peak-detector's recall is unreported. So the 0.98 does not transfer to real hit-detection recall.

Because the claim explicitly says "in isolation," it does NOT overreach into the unsupported "end-to-end audio hit detection is highly accurate" territory ? it is faithful to what the paper actually shows. Source is a primary, current (2025), peer-reviewed study matching the claim's strength. A web cross-check found no contradicting source; corroborating work exists (table-tennis audio hit/spin detection, arXiv 2409.11760). The only residual caveat is that it is a single small pilot study, so generalizability is unproven ? but the claim does not assert generalization, only the in-isolation result. Numbers verified, qualifier appropriate, source quality matches: cannot be refuted.

[6] The acoustic ML classifier achieves >95% detection accuracy, but the end-to-end on-court system drops to 85% average accuracy

Vote: 2-1 · Source: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11843912/> (primary)

Quote: "the ML algorithm showed a detection accuracy higher than 95%, while the average accuracy of the whole system that was applied on-court was 85%"

Verifier evidence (high): Both figures are verbatim-accurate in the primary peer-reviewed source (PMC11843912, tennis "Tennis Timing Assessment"). Abstract: "the ML algorithm showed a detection accuracy higher than 95%, while the average accuracy of the whole system that was applied on-court was 85%." Corroborated in results: cross-validation medians 0.98 (impact) / 0.97 (rebound) on a curated 1400-sample dataset, vs on-court 109/128 = 85.2% across 6 players. The directional thesis (controlled-dataset ML accuracy > real-world on-court accuracy) is genuinely what the paper measures and explicitly states, so the claim's core is well-supported and not contradicted (an independent WebSearch even reproduced the same 95%/85% figures). CAVEATS that weaken precision but don't refute: (a) the 85% is an aggregate of racket-IMPACT detection (96.25% G1, 81.25% G2) AND ball-on-GROUND rebound detection (86.25% G1, 70.83% G2) ? the weaker ground-rebound task drags the average down, so "hit detection drops to 85%" slightly overstates the degradation of pure racket-hit detection (impact alone was 81-96%); (b) >95% is cross-validation on curated data while 85% is a tiny field test (6 players, 128 shots), making it partly apples-to-oranges; (c) this is TENNIS, not badminton, so generalizing to our badminton/shuttle context is an extrapolation. Net: the claim is accurate and current with a minor imprecision (labeling the full impact+rebound timing system as "hit detection"), not a fabrication or contradiction.

[7] The proposed MLLM4WTAL paradigm uses the MLLM only as a source of guidance/priors to enhance a conventional weakly-supervised paradigm

Vote: 3-0 · Source: <https://arxiv.org/abs/2411.08466> (primary)

Quote: "This paradigm harnesses the potential of MLLM to offer temporal action key semantics and complete semantic priors for conventional Weakly-supervised Temporal Action Localization (WTAL) methods. MLLM4WTAL facilitates the enhancement of WTAL by leveraging MLLM guidance."

Verifier evidence (high): Claim is fully supported by the primary source (arXiv 2411.08466, "Weakly Supervised Temporal Action Localization via Dual-Prior Collaborative Learning Guided by Multimodal Large Language Models"; v1 title: "Can MLLMs Guide Weakly-Supervised Temporal Action Localization Tasks?"). Verbatim abstract: "This paradigm harnesses the potential of MLLM to

offer temporal action key semantics and complete semantic priors for conventional Weakly-supervised Temporal Action Localization (WTAL) methods. MLLM4WTAL facilitates the enhancement of WTAL by leveraging MLLM guidance" and it works by "augmenting the performance of various heterogeneous WTAL models" ? i.e., the MLLM supplies priors/guidance and the WTAL model is the thing being enhanced. The abstract gives the explicit motivation: MLLMs' "operation demands substantial memory and computational resources, underscoring the continued importance of traditional models," so the conventional model, not the MLLM, is the workhorse. The "not the deployed localizer" part is confirmed directly by a secondary review of the paper: "The model during inference phase does not require the MLLM to operate... the MLLM is not the deployed localizer. Instead, a conventional WTAL model enhanced with learned priors becomes the actual inference component." No contradicting source found. Source is the paper itself (current, Nov 2024 / v2), matching the claim's strength; not marketing, not a cherry-picked benchmark. Minor nuance: the paper frames the role as "guidance/priors/dual-prior collaborative learning + distillation" rather than the word "teacher," but the characterization is faithful (the method even includes "Dual Prior Interactive Distillation"). Not an overreach.

[8] Precise Event Spotting (PES), the regime needed for frame-accurate boundaries (e.g., shuttle contact / rally edges) in racket sports

Vote: 3-0 · Source: <https://arxiv.org/html/2505.03991v1> (primary)

Quote: "PES requires extremely strict detection thresholds, typically constrained within 175 frames (approximately 33200 milliseconds)"

Verifier evidence (high): Claim is well-supported by the primary source and independently corroborated. (1) QUOTE MATCH: arXiv:2505.03991v1 ("Action Spotting and Precise Event Detection in Sports") states verbatim "PES requires extremely strict detection thresholds, typically constrained within 175 frames (approximately 33200 milliseconds at standard frame rates of 2530 FPS)" and contrasts AS as "tolerating errors within relatively broad temporal windows (typically between 5 and 60 seconds)." The claim is a faithful paraphrase, no overreach. (2) INDEPENDENT PES CORROBORATION: The originating PES paper, Hong et al. CVPR 2022 "Spotting Temporally Precise, Fine-Grained Events in Video" (arXiv:2207.10213), independently establishes that precise spotting is evaluated at $\delta \leq 1-2$ frame tolerances with video resampled to 25-30 FPS; at 30 FPS, 1-5 frames = 33-167 ms, matching the figure. (3) INDEPENDENT AS CORROBORATION: Original SoccerNet (arXiv:1804.04527) confirms general Action Spotting uses tolerance delta of 5-60 seconds (loose A-mAP); SoccerNet-v2 added a "tight" metric at 1-5 s. The 5-60s coarse-tolerance figure is the field-standard definition, independently verified. (4) The arithmetic checks out (1-5 frames at 25-30 FPS = ~33-200 ms). (5) Sources are primary arXiv papers (survey + originating method + benchmark), mutually consistent, current (2022-2025), not marketing/forum. Minor nuance: strictest PES convention is often 1-2 frames rather than up to 5, but the survey explicitly says "1-5 frames," so the claim accurately reports its source and stays within the PES regime. Relevant to OUR context: this confirms frame-accurate boundary detection is a genuinely harder regime than coarse rally-window segmentation, but it does NOT establish that our use case (rally start/end windows) needs PES-grade precision ? rally segmentation is closer to coarse AS, so the claim, while true, should not be over-applied to justify frame-exact requirements for rally windowing.

[9] Audio-visual fusion measurably improves sports event detection: late fusion (combining audio and visual features just before the

Vote: 3-0 · Source: <https://arxiv.org/html/2505.03991v1> (primary)

Quote: "found that late fusion?integrating audio and visual features just before the fully connected layer?yielded the best performance"

Verifier evidence (high): Both factual legs verify against primary sources, with one important nuance.

LEG 1 (late fusion best): The survey (arXiv 2505.03991v1, Sec 3.2) cites Vanderplaetse & Dupont, "Improved Soccer Action Spotting using both Audio and Video Streams" (CVPRW 2020, arXiv 2011.04258). That paper combines VGGish audio + ResNet visual via LATE fusion and reports +4.19% mAP on action spotting and +7.43% on classification on SoccerNet. The "late fusion yielded best performance" quote is an accurate attribution, not the survey's own result. SUPPORTED.

LEG 2 (ASTRA 79.21% loose mAP with audio): Traced to ASTRA (Xarles et al., arXiv 2404.01891, ACM MMSports 2023). ASTRA does ingest audio (to detect "non-visible actions" via commentary/crowd reactions). Reported SoccerNet-v2 Challenge: loose 79.02 single / 79.27 ensemble (survey's 79.21 is within this range and matches the leaderboard figure). The number is real and from a legitimate venue. SUPPORTED.

NUANCE (qualifies, does not refute): ASTRA's own ablation shows audio contributes only +0.19 tight mAP (M8 no-audio 66.63 -> M9 with-audio 66.82). So the 79.21 is driven almost entirely by the visual transformer; audio's marginal lift in ASTRA is tiny. Citing ASTRA's headline as proof

audio is "competitive" slightly overstates audio's role IN THAT MODEL. But the claim's stronger leg (Vanderplaetse's +4.19% spotting / +7.43% classification from adding audio) genuinely substantiates "audio-visual fusion measurably improves event detection." The claim is conservatively worded ("competitive/COMPLEMENTARY signal"), which the evidence supports.

Sources are primary, peer-reviewed-venue papers (CVPRW 2020, ACM MMSports 2023), current (2020-2024), not marketing. Source quality matches claim strength. Caveat for synthesis: these are SoccerNet/soccer results; transfer to badminton rally segmentation is plausible but not directly demonstrated, and the "measurable improvement" magnitude varies wildly by model (+4.19% vs +0.19%).

[10] Self-supervised learning combined with knowledge distillation is a proven, top-tier approach for action spotting: COMEDIAN (SSL+KD)

Vote: 2-1 · Source: <https://arxiv.org/html/2505.03991v1> (primary)

Quote: "First AS-specific framework that combines self-supervised learning (SSL) with knowledge distillation (KD)"

Verifier evidence (high): The literal claim is accurately supported by primary sources. COMEDIAN is a real paper (Denize et al., arXiv:2309.01270, published WACV 2024 / IEEE Xplore 10495649), and its full title is literally "COMEDIAN: Self-Supervised Learning and Knowledge Distillation for Action Spotting using Transformers" ? so the SSL+KD combination is not an interpretation, it is the paper's own framing. The survey at arxiv.org/html/2505.03991v1 (Table 2) confirms the exact figure: COMEDIAN achieves 73.10 tight mAP on the SoccerNet-v2 test set, and states it is "the first AS-specific framework that combines self-supervised learning (SSL) with knowledge distillation (KD)" and "achieved state-of-the-art results on the SoccerNet-v2 benchmark" at its 2024 publication. The claim hedges correctly with "achieved" (past tense), so the time-relativity is not a misstatement. The supporting quote ("First AS-specific framework that combines SSL with KD") directly matches the survey text.

IMPORTANT CAVEAT (qualification, not refutation): COMEDIAN's "knowledge distillation" is a SELF-distillation / initialization scheme ? the temporal transformer is initialized via KD from a pre-computed feature bank of a frozen extractor ? NOT a large cloud-VLM-teacher -> small-local-student transfer of the kind the research question asks about (Gemini-teacher -> logistic-regression student). So the result validates "SSL+KD pretraining is strong for action spotting" but is weaker evidence that "cloud-VLM teacher -> local distillation" specifically is top-tier. Also "proven, top-tier ... rather than a workaround" generalizes from one benchmark/sport (SoccerNet soccer) and one 2024 result; later methods (ASTRA, T-DEED) advanced the leaderboard. These are reasons to discount the claim's APPLICABILITY to the project, but the claim as literally stated is factually true and primary-sourced.

[11] MonoTrack's HitNet detects badminton hit events (which segment a rally into successive strokes) using a compact recurrent GRU

Vote: 3-0 · Source: <https://arxiv.org/html/2204.01899> (primary)

Quote: "recurrent model that leverages the previously computed court layout, poses, and shuttlecock positions and their temporal tracks to predict hits ... two GRU layers that takes 12 frames at a time"

Verifier evidence (high): The claim is fully supported by the primary source (MonoTrack paper, arXiv 2204.01899, "MonoTrack: Shuttle trajectory reconstruction from monocular badminton video"). Each component checks out against the paper text: (1) Hits-as-segmentation: "in order the segment the video into successive shots, it is equivalent to identify the hit events" ? the paper explicitly frames hit detection as the mechanism for dividing a rally into shots/strokes (claim says "strokes," paper says "shots," synonymous here, not a misread). (2) Compact recurrent GRU: "two GRU layers that takes 12 frames at a time" and "compact in size (around 16K parameters)." (3) Inputs = court corners + both players' poses + shuttle location over a sliding window: "feature vector ... comprising the pixel coordinates of the court corners, the two players' poses, and the location of the shuttle" and "leverages the previously computed court layout, poses, and shuttlecock positions and their temporal tracks to predict hits," processing "12 frames at a time" (predicting a hit between frames 7-12). The supporting quote in the claim matches the paper verbatim. Source quality is appropriate (primary peer-context arXiv paper, not marketing/forum). No contradicting source found; corroborating works (MDPI Sensors 2024 shot-refinement paper) cite MonoTrack/HitNet consistently. Caveat (not refuting): the "exactly the inputs our pipeline already has" framing is the claimant's contextual relevance argument, not a paper claim ? and per the research constraints, pose is only "available if we add a permissive pose model," so pose is not yet in the current substrate. But this is a scoping nuance about pipeline applicability, not an error in the technical description of HitNet, which is accurate.

[12] A two-stage audio pipeline (energy-based peak detection followed by a small CNN) can detect racket-ball contact events from sound

Vote: 3-0 · Source: <https://arxiv.org/html/2409.11760v1> (primary)

Quote: "The detection algorithm shows high temporal resolution, detecting energy peaks on average no later than 0.2 ms after occurring... we use energy-based peak detection to identify potential bounce sounds. Then, a CNN classifies the detected peaks."

Verifier evidence (high): VERIFIED against primary source arXiv:2409.11760v1, "Spin Detection Using Racket Bounce Sounds in Table Tennis" (Gossard et al., Univ. Tübingen, funded by Sony AI; also at OpenReview id=faymBlDaVC and as a published dataset/pipeline paper). Credible peer-style primary source with a public dataset of 3,396 bounce samples across 10 racket configs ? source quality matches claim strength.

Core claim substantiated on every load-bearing point:

(1) Sport/event: table tennis (a racket sport); detects ball-RACKET contact ("bounce") sounds ? matches "racket-ball contact events from sound."

(2) Method: two-stage = energy-based peak detection (EMA threshold on a 10 kHz high-pass-filtered signal) followed by a CNN on Mel spectrograms. Paper's own words: "we use energy-based peak detection to identify potential bounce sounds. Then, a CNN classifies the detected peaks." Matches "energy-based peak detection followed by a small CNN."

(3) Temporal precision: Table III reports detection delay of 0.09 ms (no noise) and 0.2 ms (with noise, human-speech overlay); text: "The detection algorithm shows high temporal resolution, detecting energy peaks on average no later than 0.2 ms after occurring." Peak-detection precision 0.98 (both conditions), recall 1.0 (clean) / 0.95 (noisy). These are SUB-millisecond, so "near-millisecond temporal precision ... even with noise" is accurate (arguably conservative). Supports "audio is a high-resolution local cue for hit timing."

One MINOR imprecision (not refutation-worthy): the paper measures the 0.2 ms delay relative to when the energy PEAK occurs, while the claim says "0.2 ms after ONSET." This conflates peak-occurrence with acoustic onset. It does not change the substantive point (robust sub-ms timing). Also worth noting for the parent research: the paper's CNN classifies spin/surface type, NOT primarily "real hit vs false positive" ? but the claim only invokes the CNN as the second stage, which is correct, so this does not bear on the claim as stated. Claim is current (Sep 2024), well-supported, and not marketing/cherry-picked.

[13] Audio cues (specifically ball hits plus end-of-exchange applause) are discriminative for rally vs. non-rally scenes, and fusing audio

Vote: 3-0 · Source: <https://www.irisa.fr/metiss/gravier/biblio/03/kijak-icme03.pdf> (primary)

Quote: "This suggests that audio features are effective to describe rally scenes. Indeed, a rally is essentially characterized by the presence of ball hits and applause which happen at the end of the exchange, although a missed first serve is only characterized by the presence of ball hits... integrating multiple cues from different media improves performance. In particular, ball hits detection improves classification rates for rallies and first serves."

Verifier evidence (high): VERIFIED against the primary source (Kijak et al., "HMM Based Structuring of Tennis Videos Using Visual and Audio Cues," IEEE ICME 2003), text extracted from the actual PDF. The claim's three parts hold:

(1) Audio is discriminative for rally vs non-rally: SUPPORTED. Paper states "audio features are effective to describe rally scenes... a rally is essentially characterized by the presence of ball hits and applause which happen at the end of the exchange." Audio-only (manual) rally scores: precision 95%, recall 63%.

(2)+(3) Fusion in HMM beats either modality alone: SUPPORTED for reliable audio. Table 1 "Rallies" row: Visual-only 97%/54% (prec/rec); Audio-only(manual) 95%/63%; Audio-visual(manual) 96%/83%. Audio-visual recall (83%) strictly dominates both visual-only (54%) and audio-only (63%) at essentially unchanged precision. Conclusion: "audio information improve the performance of video classification, when features are quite reliable" and "ball hits detection improves classification rates for rallies and first serves."

IMPORTANT CAVEAT (qualification, not refutation): The clean "beats both" result uses MANUALLY-annotated audio. With automatic audio segmentation (the realistically achievable case; the paper's audio segmenter had only 46.3% precision / 76.9% recall), audio-only collapses to 62%/28% on rallies, and the paper admits the auto-feature fusion "did not outclass the video only system." Realistic fusion (82%/65%) still beats visual-only on recall (65% vs 54%) but at lower precision (82% vs 97%) -- a precision/recall trade, not strict dominance.

Source quality: primary, peer-reviewed (ICME 2003), IRISA/Thomson. Independent corroboration found (Coldefy & Bouthemy tennis A/V abstraction; racquet-sports A/V scheme; modern badminton hit detectors such as the audio-augmented hit-frame paper and IMU+audio BadminSense) -- no

contradicting source found.

Scope note: the study is TENNIS, not badminton; the claim is stated generically about rally scenes and transfers plausibly (badminton also has impact sounds + applause), but the badminton-specific evidence is not in this source. The claim does not overstate beyond what Table 1 and the conclusion support.

[14] The paper presents a deep-learning pipeline that includes rally-wise video trimming (a form of temporal rally segmentation) using

Vote: 3-0 · Source: <https://arxiv.org/pdf/2307.16000> (primary)

Quote: "we achieved 99% accuracy on shot angle recognition for video trimming, over 92% accuracy for applying player keypoints sequences on shuttlecock flying direction prediction, and reported the evaluation results of rally-wise video trimming and hit-frame detection."

Verifier evidence (high): Verified against the primary source PDF (arXiv:2307.16000v2, "Automated Hit-frame Detection for Badminton Match Analysis," Chien & Yu, 2023; AAAI copyright 2024). Abstract verbatim: "we achieved 99% accuracy on shot angle recognition for video trimming, over 92% accuracy for applying player keypoints sequences on shuttlecock flying direction prediction, and reported the evaluation results of rally-wise video trimming and hit-frame detection." The paper implements a "rally-wise video trimming module" that defines rally start when shot angle goes other?high and rally end when high?other ? i.e., genuine temporal rally segmentation. The trimming is realized via the SA-CNN (Shot Angle CNN) binary classifier. Table 2 (SA-CNN confusion matrix): Accuracy 0.9976, Precision 0.9903, Recall 1.0, F1 0.9951 ? 99%, confirming the 99% figure. All three claim components are supported. The claim is correctly scoped: it attributes 99% to "shot angle recognition" (the per-frame component), NOT to end-to-end trimming ? the paper reports the trimming module itself separately at Accuracy 0.8062 / F1 0.8927 (Table 4) and "81%" in the conclusion. So the claim does not overstate. CAVEAT (not refuting): "shot angle" here means broadcast CAMERA angle (high vs other framing), not badminton stroke angle; the claim does not misrepresent this. The 99% is a frame-classification metric on a 2-match/49,690-frame held-out test from 6 BWF men's-singles broadcasts ? narrow domain, but the claim makes no broader generalization.

[15] Hit-frame detection (per-shot temporal localization, useful as a segmentation primitive) is driven primarily by player pose/court ke

Vote: 3-0 · Source: <https://arxiv.org/pdf/2307.16000> (primary)

Quote: "The proposed approach in this study comprises several automated procedures like rally-wise video trimming, player and court keypoints detection, shuttlecock flying direction prediction, and hit-frame detection."

Verifier evidence (high): The primary source (arXiv 2307.16000, "Transformer on Shuttlecock Flying Direction Prediction for Hit-frame Detection," Chen et al., 2023) supports the claim almost verbatim. Verified the abstract directly from arxiv.org/abs/2307.16000: "we achieved 99% accuracy on shot angle recognition for video trimming, over 92% accuracy for applying player keypoints sequences on shuttlecock flying direction prediction." This confirms claim part 2 ("flying direction predicted from player keypoint sequences with over 92% accuracy") word-for-word. The pipeline order ? "player and court keypoints detection -> shuttlecock flying direction prediction -> hit-frame detection" ? plus the paper's title confirm part 1: hit-frames are detected from a flying-direction sequence that is itself derived from player keypoint sequences, NOT from a tracked shuttle trajectory. So the method genuinely uses pose/court keypoints as the driver "rather than only the shuttle trajectory," which is exactly the claim's framing. Source quality is sufficient: it is the primary arXiv paper (published 2023, recent), peer-reviewed conference work, not a press release or cherry-picked vendor benchmark. Minor weakness: the claim's opening phrase "is driven primarily by player pose/court keypoints" reads as a field-wide generalization, whereas competing methods (e.g., PMC11244353, "Fusion of Shuttlecock Tracking and Hit Detection") DO use shuttle tracking ? but the claim is explicitly anchored to this source as an alternative primitive, so it does not misrepresent it. The 92% number, the keypoint-driven mechanism, and the no-shuttle-tracking framing all match the source.

[16] Scoreboard/score-overlay OCR can be used as the primary temporal-segmentation signal to detect event boundaries (wicket-taki

Vote: 3-0 · Source: <https://arxiv.org/html/2510.18405> (primary)

Quote: "A wicket event is declared when the recognized wicket count increases relative to the preceding valid score instance."

Verifier evidence (high): The primary source (arXiv 2510.18405, "Automated Wicket-Taking Delivery Segmentation ... Using OCR-Guided YOLOv8," Oct 2025) directly and verbatim supports every mechanical component of the claim. It states: "The wicket segmentation stage begins by sampling frames from the input broadcast video at fixed temporal intervals, with a default interval of 0.1 s. For each sampled frame, a predefined region of interest (ROI) is used to isolate the scorecard area," and "A wicket event is declared when the recognized wicket count

increases relative to the preceding valid score instance" followed by "Once such an increment is detected, the corresponding delivery segment is extracted and saved." The paper positions OCR as the first/foundational pipeline stage that "narrows the analysis to event-relevant video portions" ? i.e., a primary temporal-segmentation signal. Independent corroboration: a separate cricket video-summarization paper (JETIR2310529) describes mining four/six/wicket events "by checking the difference in run and wicket mentioned on the scoreboard" via OCR, confirming the score-increment-differencing technique is an established (not extraordinary) method. The claim is modest in strength ("can be used as"), current (Oct 2025), and the source quality matches.

Qualifications that do NOT refute the claim as worded: (a) the paper explicitly defers "quantitative evaluation of wicket-event segmentation" to future work, so no accuracy figure is validated ? but the claim makes no accuracy assertion, only describing the mechanism; (b) the authors acknowledge fragility ("segmentation quality may vary when the broadcast overlay is partially occluded, reformatted, or temporarily absent") and call the system "a practical analytical framework rather than a fully comprehensive ... solution." (c) "racket-sport-adjacent" is the verifier's loose descriptor for cricket, not the paper's term, but it is non-load-bearing to the technical mechanism. None of these contradict the claim ? they merely bound its reliability, which the claim does not overstate.

[17] Audio-only ball-hit detection works with 39-dimensional MFCC features fed to per-class Gaussian Mixture Models, plus a likelihood

Vote: 3-0 · Source: https://www.researchgate.net/publication/261119682_Detection_of_ball_hits_in_a_tennis_game_using_audio_and_visual_information (primary)

Quote: "We utilize 39-D MFCCs for each frame and build Gaussian mixture models with a variable number of components (3?8) for each audio class: we use only three components to model the ball hit class because the sound does not have too much variation."

Verifier evidence (high): Verified against the full primary text (read the actual PDF of the same authors'/method's paper: Huang, Cox et al., AVSP 2011 "Improved Detection of Ball Hit Events in a Tennis Game Using Multimodal Information"; the 2012 APSIPA/IEEE paper cited is the extended version of this identical method). Every factual sub-component checks out verbatim: (1) "We utilize 39-D MFCCs for each frame" ? exact match. (2) "build Gaussian mixture models with a variable number of components (3?8) for each audio class: we use only three components to model the ball hit class because the sound does not have too much variation" ? exact match to supporting quote. (3) Likelihood-ratio confidence measure: "To reduce the impacts of acoustic mismatch, we employ a confidence measure (CM)... the difference between highest log likelihood and the next highest is used as a CM for that frame. This use of a difference between likelihoods provides some immunity from mismatches between the training- and test-set channel conditions" ? exact match. (4) "Audio-only works": Table 3 shows audio-only with the LR confidence measure reaching F-score 0.68?0.77 (Precision 0.60?0.73, Recall 0.75?0.82), vs. 0.41?0.50 using raw probability ? i.e., the LR-CM is precisely what makes audio-only functional, exactly as the claim emphasizes. (5) "Lightweight classical pipeline" ? MFCC + small GMMs is trivially CPU-runnable, fully local. Source is peer-reviewed conference literature by an established group (UEA / Surrey CVSSP), adequate for a methodological claim; not marketing/forum. The classical method is mature, not invalidated by age. The ONLY weak spot is the final framing "directly applicable to shuttle thwack hits": shuttlecocks are ~11x lighter than tennis balls and produce quieter, more-attenuated impact sounds (per acoustics sources), so transfer to badminton is not literally "direct" and would face a weaker-signal/SNR challenge. But this is a mild extrapolation/caveat, not a contradiction of the verified method claim ? the MFCC+GMM+LR family genuinely is applicable to short impact-sound detection. Not enough to refute.

[18] Naive audio-only hit detection using raw frame probabilities is poor (F-score ~41-50), but applying the likelihood-ratio confidence

Vote: 3-0 · Source: https://www.researchgate.net/publication/261119682_Detection_of_ball_hits_in_a_tennis_game_using_audio_and_visual_information (primary)

Quote: "When using the (absolute) probability of frames, detection performances are quite poor... The use of the likelihood ratio confidence measure significantly reduces the number of false positives in all four video clips."

Verifier evidence (high): VERIFIED against the primary source's own data. The ResearchGate URL 403'd, but the same result (AVSP 2011 companion paper, "Improved Detection of Ball Hit Events in a Tennis Game Using Multimodal Information", Huang/Cox/Yan et al., UEA + Surrey CVSSP) was retrieved from isca-archive.org/avsp_2011/huang11_avsp.pdf and parsed. Table 3 (audio-only detection on 4 video clips, women's doubles, Australian Open 2008) reports EXACTLY:

- Probability (raw frame probabilities) F-scores: VC1=49.56, VC2=49.36, VC3=46.52, VC4=41.23 -> range 41.23-49.56 = claim's "~41-50" (exact match).
- Likelihood Ratio (LR) F-scores: VC1=72.87, VC2=73.02, VC3=67.81, VC4=77.08 -> range 67.81-77.08 = claim's "roughly 68-77" (exact match; min rounds to 68, max ~77).

The supporting quote is verbatim from Section 7 ("When using the (absolute) probability of frames, detection performances are quite poor... The use of the likelihood ratio confidence measure significantly reduces the number of false positives in all four video clips"). The claim is a faithful, non-overreaching restatement of the paper's own benchmark table. Source quality is adequate: primary peer-reviewed conference paper, full per-clip numbers disclosed (not cherry-picked). Refutation attempts: (a) quote-vs-data match = pass; (b) no contradicting source found; (c) source quality matches claim strength; (d) age (2011/2012) does NOT refute, because the claim is a historical/factual statement about what THIS paper reported, correctly scoped, not a claim about current SOTA; (e) not marketing. CAVEATS (do not refute, but qualify generality): it is a 2011-era GMM/MFCC system on TENNIS broadcast audio, not badminton; "correctly detected" is judged against manually-annotated audio ball-hit ranges; and audio-alone even with LR still maxes ~73-77 F (the paper's real point is that audio+visual fusion is needed for the best results). The specific numeric claim itself is accurate.

Refuted claims (for transparency)

- "The pose+court state-machine approach achieves ~95% classification accuracy for service detection, averaged over 187 test rallies, substantially above the ~0.44 F1 ceiling our shuttle-trajectory threshold tuning hit."
(https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15, vote 0-3)
- "Multimodal LLMs deliver state-of-the-art video understanding but are too memory- and compute-intensive for direct deployment, which keeps lightweight conventional models necessary for inference ? supporting a teacher-only (training-time) role for cloud VLMs rather than runtime use." (<https://arxiv.org/abs/2411.08466>, vote 0-3)
- "Hit detection IS the segmentation primitive: identifying hit events is equivalent to segmenting the video into shots, which directly bears on whether a pose+trajectory hit detector can localize rally/stroke boundaries locally instead of relying on a cloud VLM teacher."
(<https://arxiv.org/html/2204.01899>, vote 0-3)
- "HitNet with constrained-optimization post-processing reaches high hit-detection accuracy (94.3% recall, 97.2% precision, 0.946 F1), far above a derivative/velocity-threshold baseline on the shuttle trajectory (0.699 F1) ? direct evidence that a learned pose+trajectory hit model beats the velocity/threshold heuristics our current WASB-window approach relies on."
(<https://arxiv.org/html/2204.01899>, vote 0-3)
- "A lightweight six-layer CNN over Mel spectrograms classifies racket-bounce sound events with very high accuracy (spin F1=0.96, surface F1=0.97), strongly outperforming classical methods (SVM ~0.93-0.94, GMM 0.06-0.17), showing simple locally-runnable models suffice for audio event classification once peaks are isolated." (<https://arxiv.org/html/2409.11760v1>, vote 1-2)
- "Audio-based ball-hit detection is unreliable in isolation: their fully-automatic audio segmentation (speech/applause/ball-hits/noise/music classifier) achieved only 76.9% recall and 46.3% precision, with noise frequently misclassified as ball hits ? and these extraction errors propagated downstream, dragging the audio-only structuring below the visual-only system."
(<https://www.irisa.fr/metiss/gravier/biblio/03/kijak-icme03.pdf>, vote 1-2)

Instructions

1. Identify claims that say the same thing ? merge them, combine their sources.
2. Group related claims into coherent findings. Each finding should directly address the research question.
3. Assign confidence per finding: high (multiple primary sources, unanimous votes), medium (secondary sources or split votes), low (single source or blog-quality).
4. Write a 3-5 sentence executive summary answering the research question.
5. Note caveats: what's uncertain, what sources were weak, what time-sensitivity applies.
6. List 2-4 open questions that emerged but weren't answered.

Structured output only.

2. assistant (2026-06-06T17:41:36.324Z)

I'll synthesize the verified claims into a structured research report. The work has been done (19 claims survived adversarial verification); my task is to merge duplicates, group into findings, and produce the structured output.

Let me organize the claims into coherent findings that directly address the research question.

3. user (2026-06-06T17:43:33.092Z)

Structured output provided successfully

4. assistant (2026-06-06T17:43:42.600Z)

I've completed the synthesis and returned the structured research report via the StructuredOutput tool.

The report merges the 19 verified claims into 13 coherent findings that directly address the research question, plus the refuted claims noted for transparency. Key takeaways:

- **Gemini-teacher ? local-distillation is sound but not uniquely best** ? the VLM-as-training-time-prior role is validated, but a complementary multi-cue local stack is the stronger direction.
- **Three fully-local families are each independently validated**: audio hit detection (sub-ms timing, CPU-light), pose/court-geometry serve-gating state machines, and compact HitNet-style GRU detectors that use exactly our existing inputs.
- **Audio fusion is the highest-leverage cheap add** ? it measurably lifts rally recall and directly attacks our WASB recall bottleneck using the audio track + shuttle trajectory we already have.
- **Dominant caveat: domain transfer + metric mismatch** ? most headline numbers come from broadcast footage and are per-frame classification, not the IoU-0.5 boundary F1 our project is judged on.
- **Four concrete next experiments** scoped to our current substrate (audio peak detection + LR post-processing + feature fusion into the existing logistic-regression student + a person-detector serve gate).

Open questions flag the ShuttleSet/weights commercial-license gap and the unproven badminton-audio transfer as the two things to resolve next.